

The Whitepaper, Translated

A restatement of *The Distributed AI Economy* (Revision 9) in the vocabulary of two ledgers: `definitions.md` — P1-P7, D1-D20, the machinery, with `examples.py` as its executable companion — and `organisational-definitions.md` — Q1-Q5, O1-O9: persons, obligations, history. Entries are cited as `D9[model]`, `Q2[attribution]`, `O2[owner]`; the paper's terms are capitalized (Frame, Cog, Op) and are always the paper's, never the ledgers'.

What “translated” means here. Every load-bearing term of the paper is restated in the vocabulary of the two ledgers — mechanical where the machinery carries it, organisational where its content is persons, obligations, or history — or what neither holds is recorded with the reason. The line between mechanical and organisational is drawn by construction — a term sits where a definition could be built under the ledgers' rules — so where the line falls is a result, not an opinion.

1. The dictionary

the paper says	the mechanical ledger	the organisational ledger, or what neither holds
model	D9[model]	—
weights, “open weights”	part of a D10[model specification]	a specification determines a set of models, never one
tokenizer, tokens	D3[tokenizer], D4[token], D6[token alphabet]	the numbering and the decoder, both deferred with triggers
context, context window	a text over A with bound L; the window is L	“context” in the paper spans the carried texts, the assembled input, and the bound — three different things
prompt	a text in the sequence given to D11[assembly]	nothing mechanically separates it from a Frame's prose
generation, “the loop”	D13[model step], D14[model run]	the ledger's names guarantee less: a model is consulted; nothing requires the sampled symbol to survive the assembly
agent harness	the Given of D14[model run]: sampler, assembly, draws, stopping	—
temperature, seeds, sampling	D12[sampler] and its draws	randomness is in where draws come from; the functions are deterministic
Guard	D15[check]; shipped: D18[check specification]	version: O4[version]; owner: O2[owner]; when it runs belongs to the caller
Gate	D16[decision]	nothing mechanical distinguishes a verdict alphabet from an action alphabet
Track	D17[run specification]	pastness: O4[version]; keeping: O9[retention]; approvals: O8[approval]; cannot pin the model; verdicts deferred
Validation Strategy	D18[check specification]	the pre-flight/in-flight/post-run placement — a property of the caller
Frame	D19[frame]	scope: O6[scope]; owner: O2[owner]; versioning: O3[history], O4[version]; inheritance: O6 Notes; shareability and discoverability are conventions

the paper says	the mechanical ledger	the organisational ledger, or what neither holds
Op	D20[checked run]	human roles: Q4[obligation], O8[approval]; tools, integration and the manifest format stay outside
Cog	D14[model run] plus carried texts	permissions: Q5[permission]; identity: Q3[succession]; tools stay outside
agent (“a Cog engaged through an Op”)	D20 with a starting text	identity: a history with attributions (Q3[succession], Q2[attribution]); memory: texts or nothing (D14 Notes)
Organizational Memory	a source of texts for the assembly’s sequence	retention: O9[retention]; access: Q5[permission]; governance and forgetting stay outside
Intelligence Hub	the site that picks which member of each specification’s set runs (D10[model specification])	custodian: keeps the carriers, seats the bindings — its registry rows evidence Q2[attribution], its access controls are Q5[permission] in carrier form
Nebari	a catalogue of specifications — texts determining sets of components	shared reading conventions for assembling Hubs; its packages have owners and histories (O2[owner], O3[history])
Nebi	transport for specifications between Hubs; pinning shrinks the determined set, never to one (D10[model specification])	the packaging format is a reading convention, codified; install logs are succession carriers (Q3[succession])
Desktop/Web App	where starting texts enter and outputs return	the person-machinery surface: a click becomes an O8[approval], obligations reach the obliged (Q4[obligation]), scopes are experienced (O6[scope])
marketplace, provenance	O7[provenance]: someone stands behind each version of a history	records of it are exchanged; the facts are bindings — see §4

2. The paper’s claims, restated

Each claim is quoted from Revision 9, then read through the theory. “Sharpened” means the theory says something more precise than the paper; “corrected” means the paper’s sentence, read mechanically, says something the machinery cannot deliver.

2.1 “behaves identically (within generative AI limits)” — §3.2, sharpened

The parenthesis is doing all the work, and the theory states exactly what it contains. By D14[model run], a run is fixed by four things: the three components, the starting text, and the draws. A shipped artifact can pin the starting text and the draws exactly, since both are texts. It cannot pin the model: by D10[model specification], a text determines a *set* of models, and the set is a singleton only if the number set and evaluation order are pinned too — which no file carries, because they are properties of the machine (Clarification: hardware arithmetic is exact; two evaluation orders are two functions, hence two models). So “identical (within generative AI limits)” translates to: *identical starting text and draws, some member of the same set of models*. The same limit applies to Guards (§2.7 below), which the paper does not flag.

2.2 “A Frame is not a prompt” — §4.3, half corrected, half sharpened

Mechanically, the prose part of a Frame *is* a prompt. A text arrives at D11[assembly] carrying its symbols and nothing else — no provenance, no authority — and for a

concatenating assembly, carrying `f` and receiving `t` produces the same input as receiving `f` followed by `t` typed by hand. Installed and typed are extensionally the same.

What makes the sentence true is the paper's own §4.3 heading: *"One of the critical things that Frames can do is define a Validation or Verification tool (a Guard) that must be called and pass."* That is D19[frame]: a check specification that is itself a text over `A` — the one artifact that both enters the model's input and declares what checks the output. The declaration is read outside the loop, by the caller; that is what a prompt does not have.

Consequence the paper should want: a Frame that declares no Guard has no mechanical content — it is a text like any other, and every one of its §4.3 properties is organisational: scoped is O6[scope], owned is O2[owner], versioned is O3[history] with O4[version], inheritable is O6's Notes; shareable and discoverable are conventions. The paper's Frames divide into two kinds, and only one kind does anything the machinery can see.

2.3 Frames "orient" Cogs; the Cog "must respect" them — §4.3-4.4, corrected

A text cannot bind a model. D8[scoring] is total on `A` — every symbol receives a number whatever the input says — so a text can shift every score and cannot remove a symbol from `A`; exclusion, where it happens, is the sampler's act (D12), not the text's. "The Cog must follow the Frame" is not a sentence the machinery can make true: obedience is score-mediated bias, not constraint. Constraint exists in the architecture — it lives in D15[check] and D20[checked run], outside the loop, where a failing output is refused or retried. This is not only a criticism: it is the strongest available argument for why Guards must exist as separate components. If Frames could bind, Guards would be redundant.

2.4 Boundaries, and prompt injection — §5.8, derived

D11[assembly] takes a *sequence* of texts and returns *one* text. A sequence records where each text begins and ends; a single text does not. So any distinction among the assembled texts — system against user, policy against data — survives only as delimiter symbols written into the content, and content can contain those symbols. Prompt injection is therefore not an implementation bug but a corollary of the assembly's type. The paper cites OWASP (§5.8); the theory derives the vulnerability from one line: `asm :`
 $(A^*)^* \rightarrow A^{\leq L}$.

The same type explains why a Consensus Guard must receive outputs *before* assembly: in the example system, `{aba'}`-style sequences with different agreement verdicts assemble to the same text, so no check on the assembled text can compute agreement (`examples.py` asserts this). D15[check] takes a sequence of texts for exactly this reason.

2.5 "The harness is the loop; a Cog is what the loop ships as" — vocabulary table, confirmed

The paper's sentence translates exactly. The loop is D13[model step] iterated as D14[model run]; what ships is the Given — model, sampler, assembly, carried texts — plus what the ledger cannot express (tools, permissions). The three Cog types of §4.4 (model-heavy, context-heavy, combined) are three ways of weighting the same Given: which of `M` and the carried texts dominates the artifact. That is packaging, not structure — the theory sees one kind of thing shipped three ways.

“A well-constructed Cog is, in large part, a well-managed context” (§4.4) translates to: the Cog’s useful degrees of freedom are the assembly and the carried texts — which is where D11[assembly]’s Notes place everything that matters: what is kept, what is dropped, and what order survives.

2.6 “Guards check. Gates decide.” — §5.2, confirmed, then reduced

Confirmed as types: a Guard is $\text{chk} : (A^*)^* \rightarrow V$, a Gate is $\text{dec} : V^* \rightarrow C$ — many things reduced to one symbol, twice over, at two levels. The reduction the paper does not state: nothing mechanical distinguishes V from C . Which alphabet holds verdicts and which holds actions is fixed by position in the chain, not by any property of the alphabets.

The seven Gate actions of §5.2 — continue, pause, request human approval, escalate to an expert, retry with a different Cog, run more validation, stop — reduce to **three mechanical roles** (D20[checked run]): *continue* is the accept (the value stands), *stop* is the refuse (no value, final), and the other five are *the next level* — another round, differing only in who or what acts before it, a difference the function cannot see. The five are real, but they are organisational: each names whose obligation the action triggers — pause the operator’s, request-approval the approver’s (its artifact O8[approval]), escalation the expert’s — Q4[obligation] doing the work the names suggest. In the example system this is computed: *review* does nothing except let the recursion continue.

2.7 Guards “deterministic ... probabilistic ... human review” — §5.1, corrected

Three different things are being called one thing:

- A deterministic Guard is a check — but a *shipped* Guard is a D18[check specification], not a check, for the same two reasons a shipped model is a specification: it may use a model (§5.7 says some Guards are Cogs), and even a model-free implementation is underdetermined by evaluation order. **The marketplace’s unit of exchange is a specification.** What §6.1 says of Frames — valuable “because of who stands behind it, not because of its code” — is therefore true of Guards for a mechanical reason, not only a commercial one.
- A probabilistic Guard is a deterministic check consuming a draw, exactly as D12[sampler] consumes one. Declaring the draw turns “probabilistic” from an unstateable property into a recordable input.
- A human is not a check at all. By P4[function], a check gives one verdict per input; a person does not. An Expert Guard is a *slot where a verdict enters from outside* — an input, like a draw, not a function. This is the honest mechanical reading of “human-in-the-loop”: the human is data the Op consumes, and everything about *who* may supply that data is organisational: Q5[permission] says who may, and O8[approval] is the artifact supplied.

Regression and Drift Guards (§5.5) translate cleanly and usefully: a check’s input is a sequence of texts, and nothing requires those texts to come from one run — a drift check is a check over outputs of many runs. The *time* part (which runs, how sampled) is organisational; the comparison is D15[check] as it stands.

2.8 “an organization can reconstruct why a decision was made” — §5.3, corrected to *what*

A Track can hold what ran: by D17[run specification], it pins the starting text and the draws exactly, and the components only as specifications (§2.1’s limit). That reconstructs *what* happened — and, if draws are recorded, allows replay up to the choice of model within the specified set. “Why” is not in the machinery at all: a run has no state of its own (D14 Notes), nothing is carried between steps except the text, and the scoring’s numbers are the whole of the model’s contribution. The theory’s advice to the paper: claim *what*, and claim it precisely — it is a strong claim, because texts pin texts exactly. Note also that a text composed before the work and one composed after it are the same kind of object; a Track is a run specification whose pastness is being an earlier item of a history (O4[version]) and whose keeping is someone’s duty (O9[retention]).

2.9 The lifecycle — §5.4, unified

Pre-flight, in-flight, post-run, continuous: one construction at different grains. Pre-flight is a check on (t) before any round. Post-run is D20[checked run] as defined. In-flight is the same wrap applied at D13[model step] grain instead of D14[model run]. Continuous is a check whose input sequence spans runs (§2.7). The paper presents four stages; the theory presents one operation and a choice of where to apply it — which is simpler to implement and simpler to audit, since there is only one thing to get right.

2.10 “An agent, precisely, is a Cog engaged through an Op, given identity and memory by the Hub” — §4.2, translated with a hole named

The Cog-through-Op part translates: D20[checked run] with a starting text. The rest of the sentence names the residue precisely: *identity* is not mechanical — nothing in the machinery distinguishes one caller from another — but organisational: the continuing actor is a history of texts with attributions (Q3[succession], Q2[attribution]); and *memory* is texts or it is nothing — whatever an agent “remembers” within a run is in the running text (and subject to D11’s overflow); whatever it remembers across runs enters the next run as part of the starting text, supplied from outside. The paper’s claim that whoever holds context, memory, and history owns the agent is thus mechanically literal: the agent is its texts, plus functions anyone in possession of the specifications can re-instantiate.

3. What the paper is about, seen from the theory

Strip the market framing and the paper adds one structure around the generation loop, in four moves, all of which the theory holds:

1. **Declare in advance, in text, what must hold** — D18[check specification], carried by D19[frame] when the declaration travels inside model input.
2. **Check the work against the declaration** — D15[check].
3. **Decide from the verdicts** — D16[decision].
4. **Record enough to fix what ran** — D17[run specification].

And the fifth move, which makes it an economy: the declarations and checks are themselves texts, so they can be shipped. Every artifact in the paper is a file because only

texts can be exchanged, and the theory adds the caveat that gives the marketplace its real shape: **texts determine functions only up to a set** (D10, D18). What a marketplace moves is specifications; what runs in a Hub is members of the sets they determine.

The accountability plane, for all its vocabulary, is **two shapes**: many things reduced to one symbol (check, decision), and a text that determines a set of functions (model, run, and check specifications). Guards, Gates, Tracks, and Validation Strategies are four names for two operations. That is a simplification the paper's authors can adopt, or refute by naming the distinction the theory failed to see.

4. The residue, stated positively

Everything the machinery sees is a text, a function, or a number. Everything the paper adds is about *people standing behind* texts and functions: who wrote a specification, who approved a verdict's supplier, who may install which artifacts, who answers when the output is wrong. Scope, inheritance, ownership, versioning, identity, retention, approval — none of these is mechanical, all for one reason: a text arrives carrying its symbols and nothing else.

This is not the theory dismissing the paper. It is the theory locating the paper's actual thesis. §6.1 says it in the paper's own words: "*When generation is free, provenance is the product.*" Provenance is precisely what the machinery cannot represent — which is why it cannot be generated, why it cannot be faked by anything inside the loop, and why it can be the product. The paper's business case rests on the residue being real and hard, and the theory confirms both.

What the theory adds to the thesis: since provenance is organisational, **every mechanism the paper proposes for it (signing, versioning, marketplace vetting, accountable owners) must live outside the loop and be verified outside the loop**. Anything that claims to carry provenance *through* the model — a watermark in the text, a special Frame the model "respects" — is refuted by D8's totality and D11's boundary loss before it is built.

The residue's ledger is `organisational-definitions.md`, under the same rules. Its primitives are the bindings themselves: person, attribution, succession, obligation, permission. Its definitions cover group, owner, history, version, delegation, scope, approval, retention — and provenance, defined as *that someone stands behind each version of a history*: bindings twice over, which makes the closing claim above a derivation — what a marketplace moves is records claiming provenance; the records translate, the claimed facts never do.

5. Testable claims

The translation makes the paper falsifiable in specific places. For any real Hub or the reference implementations:

1. **Does the assembly distinguish installed context from typed context?** Under overflow, a position-aware assembly (always keep the first k symbols) can make a

Frame’s survival a guarantee; a plain concatenator cannot. If no assembly distinguishes them, Frame persistence is convention, not mechanism (§2.2).

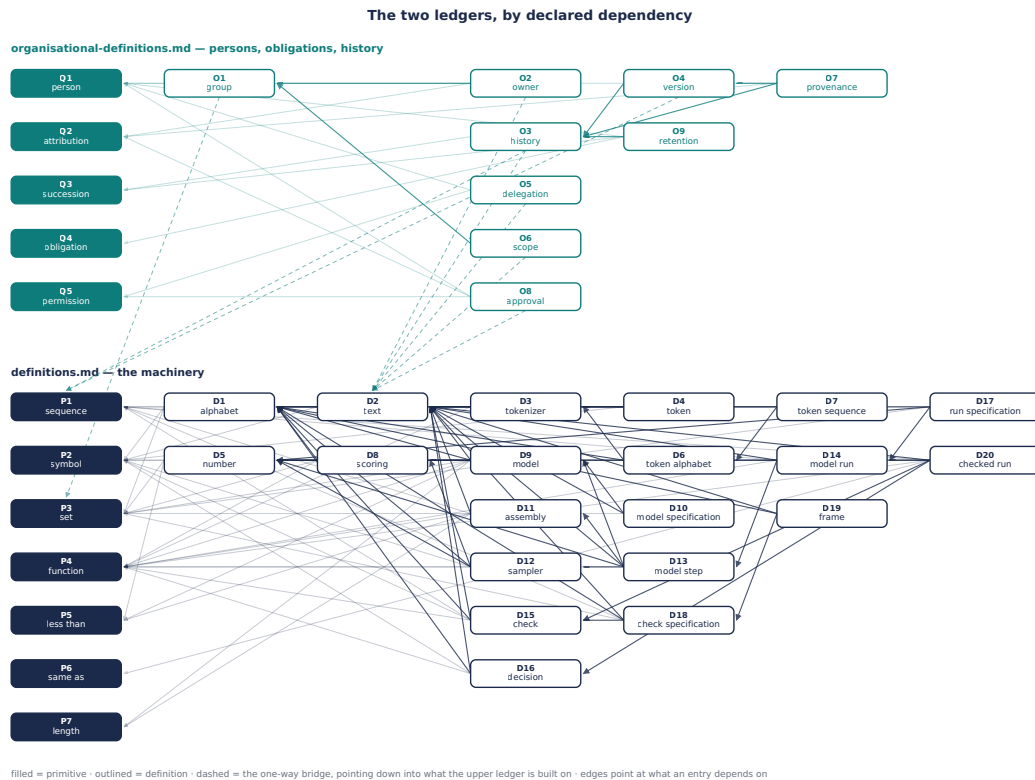
2. **What fraction of published Frames declare a Guard?** Only those have mechanical content; the rest are prose riding on organisational properties (§2.2).
3. **Are draws recorded in Tracks?** Without them, replay is impossible even within D10’s limit, and “reconstruct” means less than it appears to (§2.8).
4. **Is the numeric regime pinned anywhere** — number set, evaluation order, kernel versions? If not, “behaves identically” has no mechanical referent across Hubs (§2.1).
5. **Are Guard sequences ordered in manifests?** D18 requires the order to be declared, or the same Validation Strategy can yield different actions from the same verdicts (§2.6).
6. **Do Gate policies document who acts on each non-terminal action?** The five “next level” actions differ only there; if the *who* is undocumented, the seven-action vocabulary is decoration over three behaviours (§2.6).
7. **Where does a human verdict enter, and is it recorded as an input?** If expert review is implemented as “a Guard”, the implementation contains a non-function and will not replay (§2.7).
8. **Which classes of texts must have non-empty owners, and where is that enforced?** Ownership is partial by design — most texts have no one standing behind them, and should not — so §4.3’s *Owned* row and §6.1’s “named, accountable owner” are a policy about the governed artifacts, not a property of texts (the organisational ledger’s *owner*). The check: find the registry or equivalent carrier that makes the policy true, and the exact classes it covers.

Appendix — section map

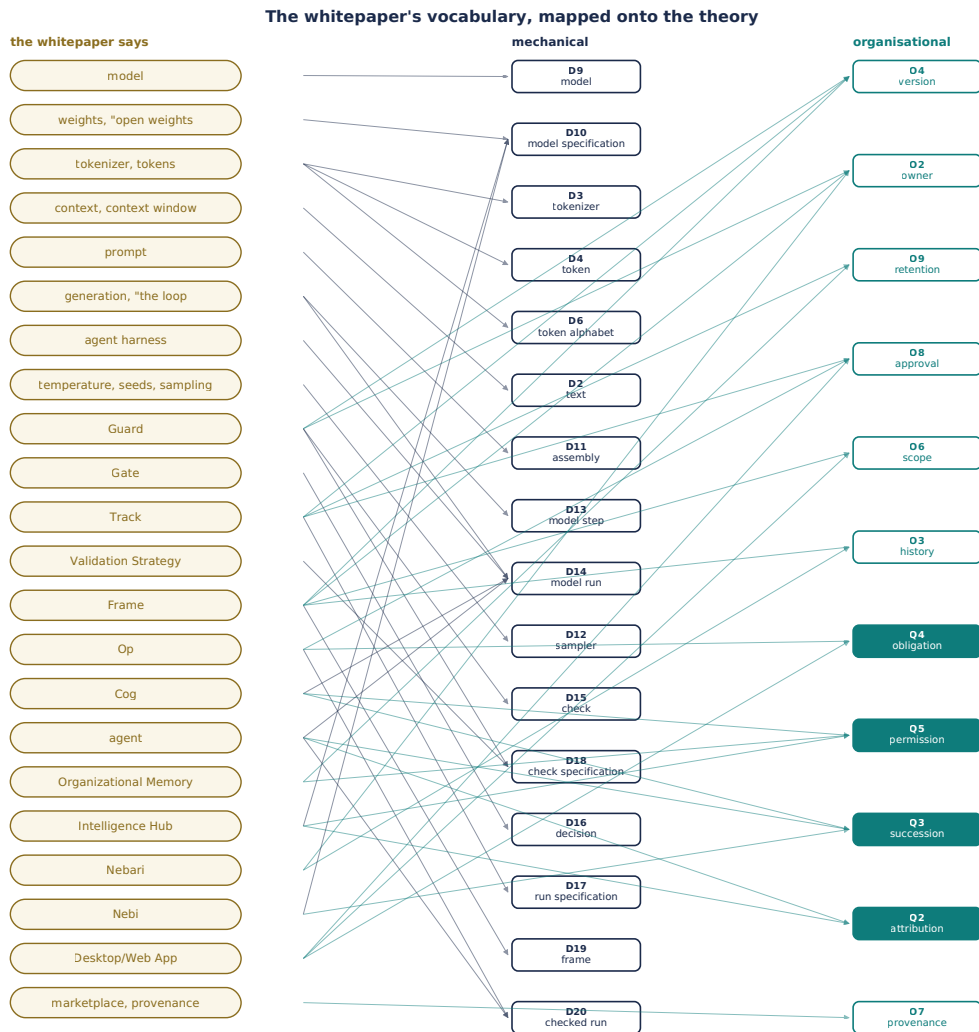
paper §	content	status
Exec. summary, 1, 2	problem, vision, market	organisational and market framing; no mechanical content to check
3.1–3.3	Nebari, Nebi, Hub	infrastructure — where functions run; §3.2’s reproducibility claim: see §2.1
3.4	Organizational Memory	a source of texts for assembly; governance is residue
4.2	Where Agents Fit	§2.10
4.3	Frames	§2.2, §2.3
4.4	Cogs	§2.5
4.5	Ops	D20[checked run]; composition of Ops not yet in the theory
5.1–5.3	Guards, Gates, Tracks	§2.6–§2.8
5.4–5.6	lifecycle, categories, strategy	§2.9, §2.7; the §5.6 manifest is a D18 with thresholds via D5/P5
5.7–5.8	validation completes / open source	§2.4; “Guards as Cogs” feeds §2.7’s specification point
6	marketplace	§3’s fifth move; §4
7–11	app, ecosystem, positioning, roadmap	organisational; no mechanical claims found

Not yet in the theory, recorded with triggers in `definitions.md`: probability distribution, boolean/predicate, the decoder, token numbering, whether zero items is a sequence, arithmetic, Op composition (§4.5's "Ops can invoke other Ops"), and a per-level record of a checked run (trigger in the mechanical ledger's log).

Figures



The two ledgers, by declared dependency



The whitepaper's vocabulary, mapped onto the theory

Generated from the ledgers and the dictionary above by `../../tools/make_graphs.py`; re-generated after any change to either.